Question				Marking details	Marks available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(b)	(i)		moles of $(COO)_2Ca$ present = $\frac{11.52}{128}$ = 0.0900 (1) 1:1 ratio therefore 0.0900 mol of H_2SO_4 needed volume needed = $\frac{0.0900}{2} \times 1000$ = 45.0 cm^3 (1)	1	1		2	1	
		(ii)		7.15g of $(COOH)_2.2H_2O$ dissolves in 50 cm^3 of solution at 20ºC (1) total moles of $(COOH)_2.2H_2O$ present is 0.0900 total mass of $(COOH)_2.2H_2O$ formed = 0.0900 × 126 = 11.34 g (1) therefore, mass crystallising out = 11.34 – 7.15 = 4.19 g (1) ecf possible from incorrect moles in (b)(i)	1	1	1	3	2	
	(c)			$3CF_2H_2 + 3(COO)_2Na_2 \rightarrow C_6F_6 + 3Na_2CO_3 + 3H_2O$ award (1) for all formulae correct award (1) balancing only if all formulae correct		2		2		